

makes the consequences of observations one-sided. It is not much more difficult than that.

This asymmetry is immensely practical. It tells us that we do not have to be complete skeptics, just semiskeptics. The subtlety of real life over the books is that, in your decision making, you need be interested only in one side of the story: if you seek *certainty* about whether the patient has cancer, not *certainty* about whether he is healthy, then you might be satisfied with negative inference, since it will supply you the certainty you seek. So we can learn a lot from data—but not as much as we expect. Sometimes a lot of data can be meaningless; at other times one single piece of information can be very meaningful. It is true that a thousand days cannot prove you right, but one day can prove you to be wrong.

The person who is credited with the promotion of this idea of one-sided semiskepticism is Sir Doktor Professor Karl Raimund Popper, who may be the only philosopher of science who is actually read and discussed by actors in the real world (though not as enthusiastically by professional philosophers). As I am writing these lines, a black-and-white picture of him is hanging on the wall of my study. It was a gift I got in Munich from the essayist Jochen Wegner, who, like me, considers Popper to be about all “we’ve got” among modern thinkers—well, almost. He writes to us, not to other philosophers. “We” are the empirical decision makers who hold that uncertainty is our discipline, and that understanding how to act under conditions of incomplete information is the highest and most urgent human pursuit.

Popper generated a large-scale theory around this asymmetry, based on a technique called “falsification” (to falsify is to prove wrong) meant to distinguish between science and nonscience, and people immediately started splitting hairs about its technicalities, even though it is not the most interesting, or the most original, of Popper’s ideas. This idea about the asymmetry of knowledge is so liked by practitioners, because it is obvious to them; it is the way they run their business. The philosopher *maudit* Charles Sanders Peirce, who, like an artist, got only posthumous respect, also came up with a version of this Black Swan solution when Popper was wearing diapers—some people even called it the Peirce-Popper approach. Popper’s far more powerful and original idea is the “open” society, one that relies on skepticism as a *modus operandi*, refusing and resisting definitive truths. He accused Plato of closing our minds, according to the arguments I described in the Prologue. But Popper’s biggest idea was his insight concerning the fundamental, severe, and incurable unpredictability of the world, and that I will leave for the chapter on prediction.*